INCH-POUND

MIL-R-39007G
AMENDMENT 1
23 March 1993

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE),
ESTABLISHED RELIABILITY
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-39007G, dated 16 May 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

TABLE II, designation and resistance column, add "R100 to R988 inclusive" and "0.1 to 0.988 inclusive" respectively.

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Add the following new paragraph:

"3.5.1.5.3 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat (see 6.9.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

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TABLE VII. Resistance temperature characteristic, delete and substitute:

Resistance range	Resistance temperature characteristic parts per million/°C
0.1 ≤ R ≤ 0.499	±650
0.499 < R ≤ 1.0	±400
1.0 < R ≤ 10	±50
10 < R	±20

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3.22.1, end of first sentence delete "±0.5 percent" and substitute "±(0.5 percent +0.005 ohm)."

3.22.2, end of first sentence delete "±1.0 percent" and substitute "±(1.0 percent +0.005 ohm)."

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Add the following new paragraph,

"3.26.4 Beryllium oxide (BeO). Manufacturers which use beryllium oxide in their construction shall mark each resistor body and resistor package with the symbol "BeO".

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4.4.4.2, add the following sentence, "In the event the contractor meets or exceeds 100 ppm for ppm-2 the qualifying activity shall take the actions specified in EIA-554, method B."

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4.6.1.2.5.2, delete the last sentence and substitute:

"If the lot fails this solderability test the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of the specification."

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4.7.3a., delete in its entirety and substitute:

"a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better."

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Add the following new paragraph:

"6.9.1 Tin plated finishes. Tin plating is prohibited (see 3.5.1.5.3) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

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6.7, add to end of last sentence, "and TABLE XVI-1 within".

Add the following new table:

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TABLE XVI-1. Performance requirements.

Style		RWR71 1/	RWR74 1/	RWR78	RWR80	RWR81	RWR84	RWR89	RWR82
Maximum resistance temperature characteristic in parts/million/°C-ppm (Reference to 25°C)	0.1 ≤ R ≤ 0.499	±650 ppm	±650 ppm	±650 ppm	±650 ppm	±650 ppm	±650 ppm	±650 ppm	±650 ppm
	0.499 < R ≤ 1.0	±400 ppm	±400 ppm	±400 ppm	±400 ppm	±400 ppm	±400 ppm	±400 ppm	±400 ppm
	1.0 < R ≤ 10	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm
	10 < R	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm
Maximum ambient temperature at rated wattage (see figure 6)		25°C	25°C	25°C	25°C	25°C	25°C	25°C	25°C
Maximum ambient temperature at zero wattage derating (see figure 6)		275°C	275°C	275°C	275°C	275°C	275°C	275°C	275°C
Power rating in watts		2	5	10	2	1	7	3	1.5
Maximum percent change in resistance ± 2/	Conditioning (see 3.8)	.2	.2	.2	.2	.2	.2	.2	.2
	Thermal shock (see 3.12)	.2	.2	.2	.2	.2	.2	.2	.2
	Low temperature storage (see 3.14)	.1	.1	.1	.1	.1	.1	.1	.1
	Short-time overload (see 3.15)	.2	.2	.2	.2	.2	.2	.2	.2
	Dielectric withstanding voltage (see 3.16)	.1	.1	.1	.1	.1	.1	.1	.1
	Moisture resistance (see 3.18)	.2	.2	.2	.2	.2	.2	.2	.2
	Shock (specified pulse) (see 3.20)	.1	.1	.1	.1	.1	.1	.1	.1
	Vibration, high frequency)see 3.21)	.1	.1	.1	.1	.1	.1	.1	.1
	Life (see 3.22)	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22
	High temperature exposure (see 3.23)	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23
	Insulation resistance (see 3.17)	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min
	Insulation resistance (after moisture resistance) (see 3.18)	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min
Resistance tolerances ± percent (see table IV)		1,.5,.1	1,.5,.1	1,.5,.1	1,.5,.1	1,.5,.1	1,.5,.1	1,.5,.1	1,.5,.1

1/ Not for use on new design; for replacement purposes only.
2/ Where total resistance change is 1 percent or less, it shall be considered as ±(___ percent +0.005 ohm).

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 80, 85
DLA - ES

User activities:
Navy - MC, CG
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1309)

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